

Arijit Dey

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EDUCATION

Degree	Specialization	Institute	CGPA	From	To
MS	Electrical and Computer Engineering	University of California, Los Angeles	3.85/4	2023	2024
M.Tech	Electrical Engineering	Indian Institute of Science	8.0/10	2021	2023
B.E.	Instrumentation and Electronics Engineering	Jadavpur University	8.91/10	2017	2021

WORK EXPERIENCE

• Data Scientist - AI Health Analytics

Aug'24 - April'26

Symviq India pvt. Ltd.

- Designed and implemented vision-language based models for the classification of Chronic Kidney Disease and Chronic Heart Disease using biomarkers from retinal fundus images. Built a foundational model with 86M parameters to support multiple downstream tasks in medical imaging.
- Trained a SimCLR-based vision encoder as part of a OpenAI CLIP model, enabling the use of textual descriptions to provide contextual embeddings for images. Achieved 91% accuracy on a proprietary chronic disease dataset, demonstrating strong model performance and generalization.
- Deployed the model using FastAPI in the backend

RELEVANT COURSES TAKEN

- **IISc:** Random Processes, Linear Algebra, Pattern Recognition and Neural Networks (Machine Learning 101), Advanced Deep Representation Learning
- **UCLA:** Convex Optimization, Neural Networks and Deep Learning, Advanced Neural Networks and Deep Learning, Decision Making in Stochastic Processes

PUBLICATIONS

- **Arijit Dey**, Bahman Ghahsifard. *Convergence of Asynchronous Stochastic Gradient Descent for Polyak-Łojasiewicz Functions³*. Submitted to *IEEE Transactions on Automatic Control*.
- **Arijit Dey**, Bahman Ghahsifard. *Graph State Networks (GSNs): Persistent Nodewise Selective State Space Models* Accepted at *Transactions on Machine Learning and Research (TMLR)*. Code available at <https://github.com/arjitcodespace/GSN> .

PROJECTS

• Graph Neural Networks (GNNs) Module

Sep'25 - Present

Personal Project

- Developed a library using TensorFlow/Keras for basic GNN training/inference.
- Supports graph convolution, graph attention, differentiable pooling, etc via its flexible API.
- Available on <https://github.com/arjitcodespace/GNN-Keras3>

• Text-to-Image Synthesis Optimization

Mar'24 - June'24

Prof. Vwani P Roychowdhury, UCLA

MS Project

- Researched text-to-image GAN architectures, optimizing CBAM under hardware constraints.
- Integrated CLIP embeddings for stronger text-conditioning, boosting semantic alignment. Achieved FID 9.85 with a 10.6M param generator vs. StyleGAN-2s 2.84 (49M params).
- Framework: PyTorch

• Implementing a Mini-GPT

Mar'24

Prof. JC Kao, UCLA

Course Project

- Implemented Mini-GPT for story generation, using weight tying to reduce parameters and increase robustness.
- Achieved Top-3 token Accuracy of 88.6% on TinyStories Dataset.
- Framework: PyTorch

¹<https://www.linkedin.com/in/arjit-d-322066bb/>

²<https://arjitcodespace.github.io/>

³<https://arjitcodespace.github.io/assets/papers/asgd.pdf>